Program

```
#include <stdio.h>
#include<stdlib.h>
#define TABLE_SIZE 10
int h[TABLE_SIZE]={NULL};
void insert()
{
 int key,index,i,flag=0,hkey;
 printf("\nenter a value to insert into hash table\n");
 scanf("%d",&key);
 hkey=key%TABLE_SIZE;
 for(i=0;i<TABLE_SIZE;i++)    {
    index=(hkey+i)%TABLE_SIZE;
    if(h[index] == NULL)     {
      h[index]=key;
       break;
    }
   }
```



```
   if(i == TABLE_SIZE)
    printf("\nelement cannot be inserted\n");
}
void search()
{
 int key,index,i,flag=0,hkey;
 printf("\nenter search element\n");
 scanf("%d",&key);
 hkey=key%TABLE_SIZE;
 for(i=0;i<TABLE_SIZE; i++)
 {
   index=(hkey+i)%TABLE_SIZE;
   if(h[index]==key)
   {
     printf("value is found at index %d",index);
     break;
   }
 }
 if(i == TABLE_SIZE)
   printf("\n value is not found\n");
}
void display() {

  int i;
  printf("\nelements in the hash table are \n");
  for(i=0;i< TABLE_SIZE; i++)
  printf("\nat index %d \t value =  %d",i,h[i]);
}
int main() {
   int opt,i;
   while(1)    {
      printf("\nPress 1. Insert\t 2. Display \t3. Search \t4.Exit \n");
      scanf("%d",&opt);
      switch(opt)       {
         case 1:
            insert();
            break;
         case 2:
            display();
            break;
         case 3:
            search();
            break;
         case 4:exit(0);
      }
   }
}
```

OUTPUT:

```
Press 1. Insert  2. Display    3. Search   4.Exit
1
enter a value to insert into hash table
34

Press 1. Insert  2. Display    3. Search      4.Exit
1

enter a value to insert into hash table
45

Press 1. Insert  2. Display    3. Search      4.Exit
2

elements in the hash table are

at index 0      value =  0
at index 1      value =  0
at index 2      value =  0
at index 3      value =  0
at index 4      value =  34
at index 5      value =  45
at index 6      value =  0
at index 7      value =  0
at index 8      value =  0
at index 9      value =  0
Press 1. Insert  2. Display    3. Search      4.Exit
3
enter search element
45
value is found at index 5
Press 1. Insert  2. Display    3. Search      4.Exit
4
```